Name: Siddharth Maru **Date: 29/08/2026**
Div: T13 **Roll No: 60**

Agile Software Development and DevOps Lab

Experiment 5

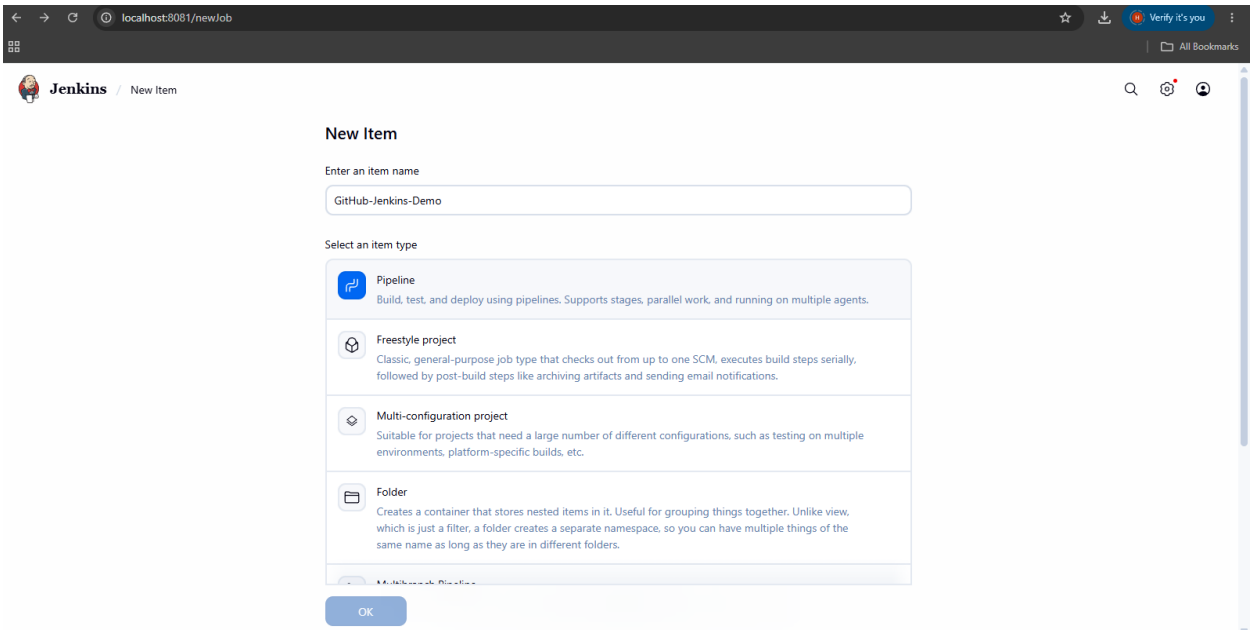
Aim: To automate application deployment using GitHub/Jenkins.

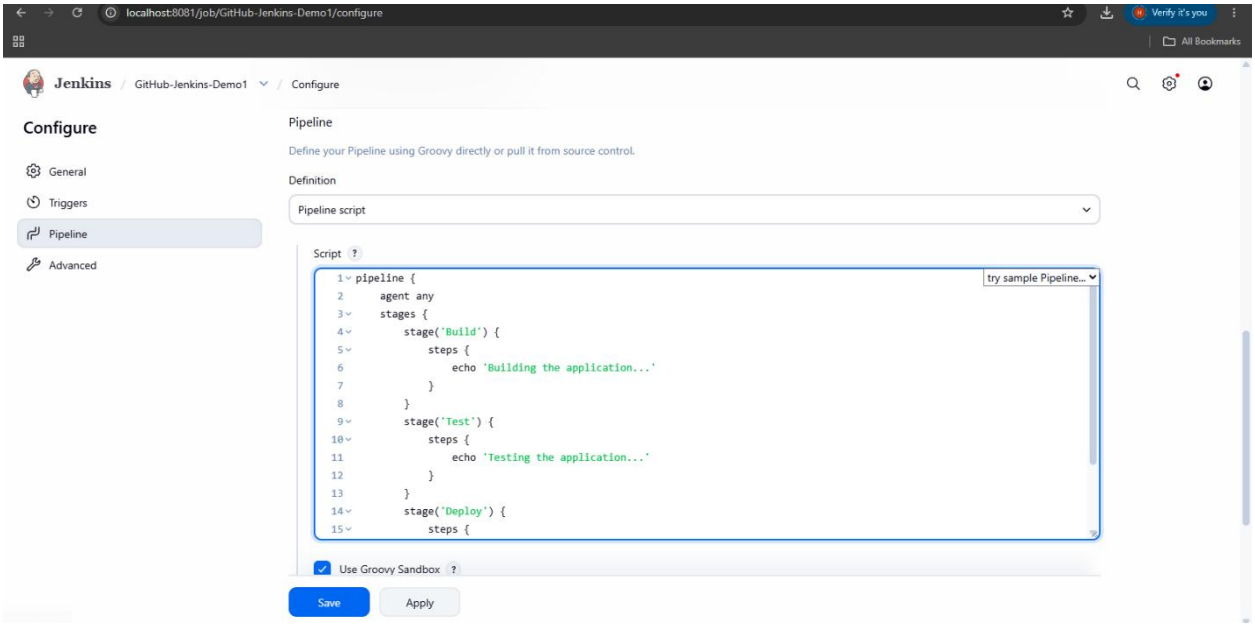
LO mapped: LO3

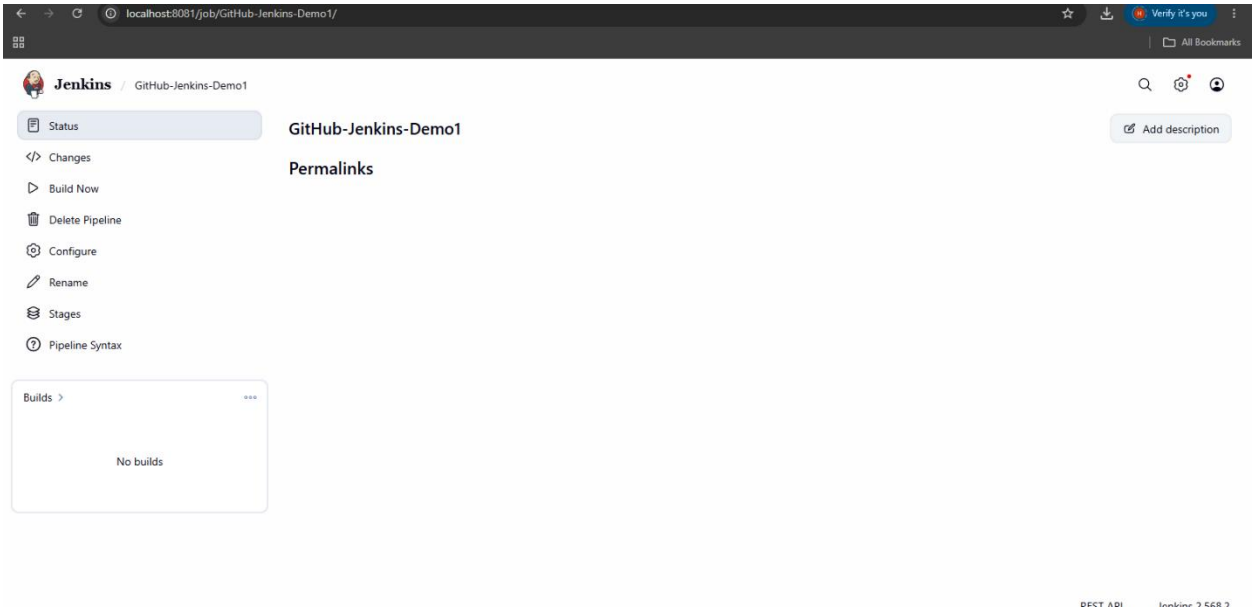
LO: Build and manage automated CI/CD pipelines for application development, testing, packaging and deployment.

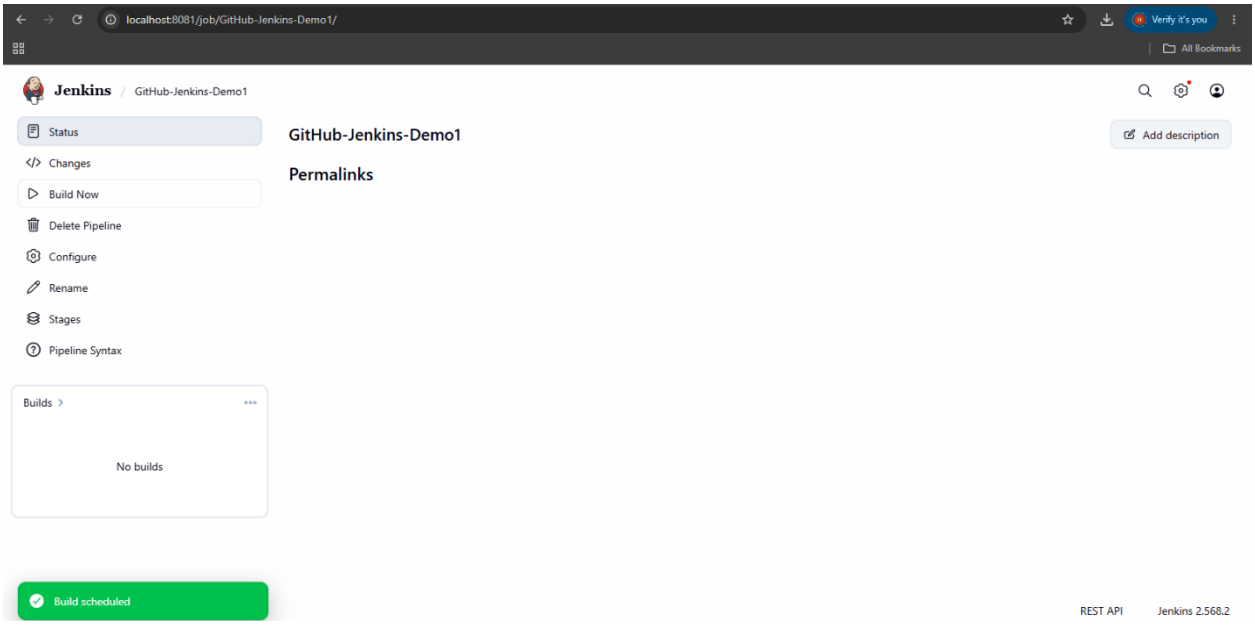
Theory:

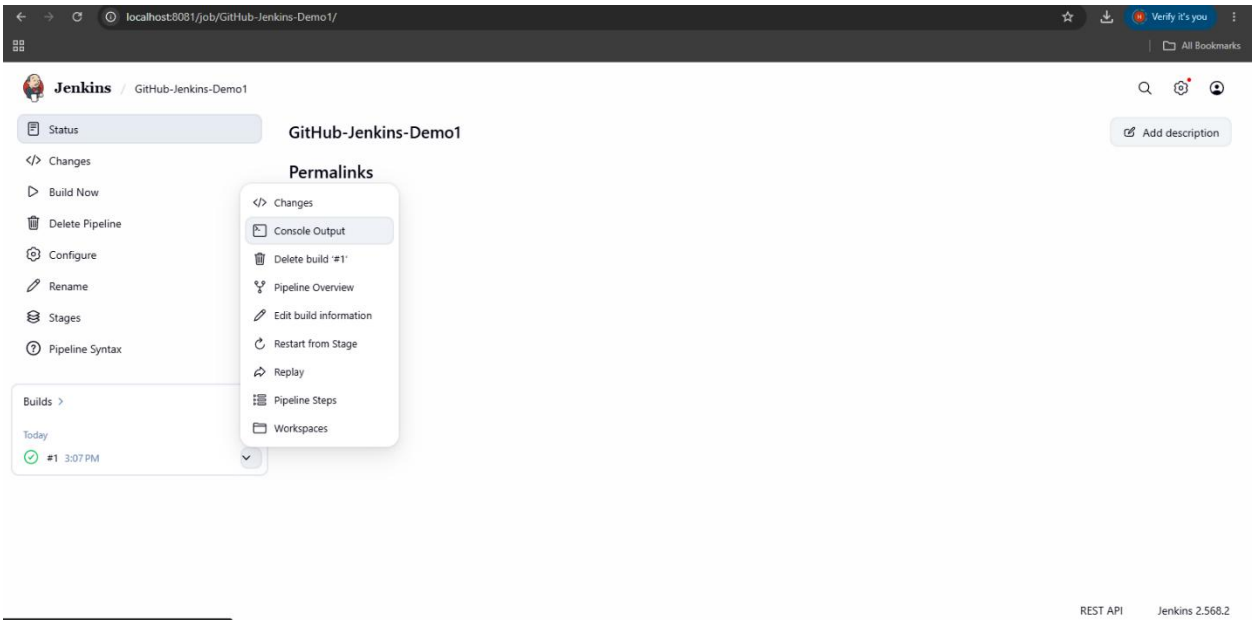
Output:

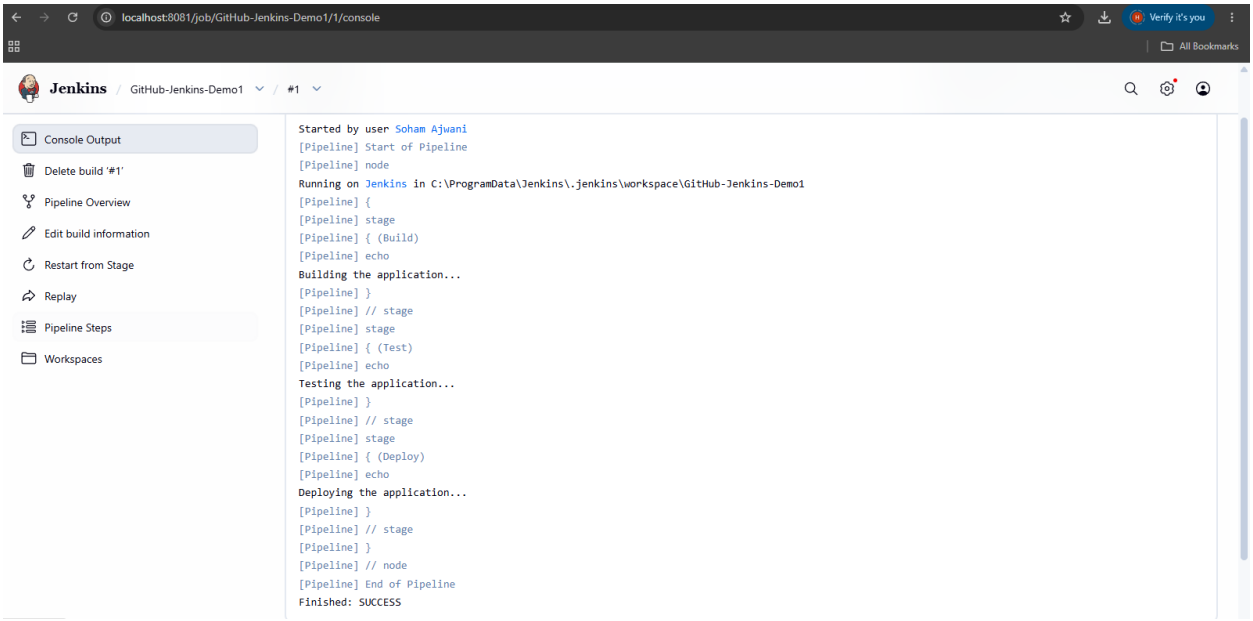


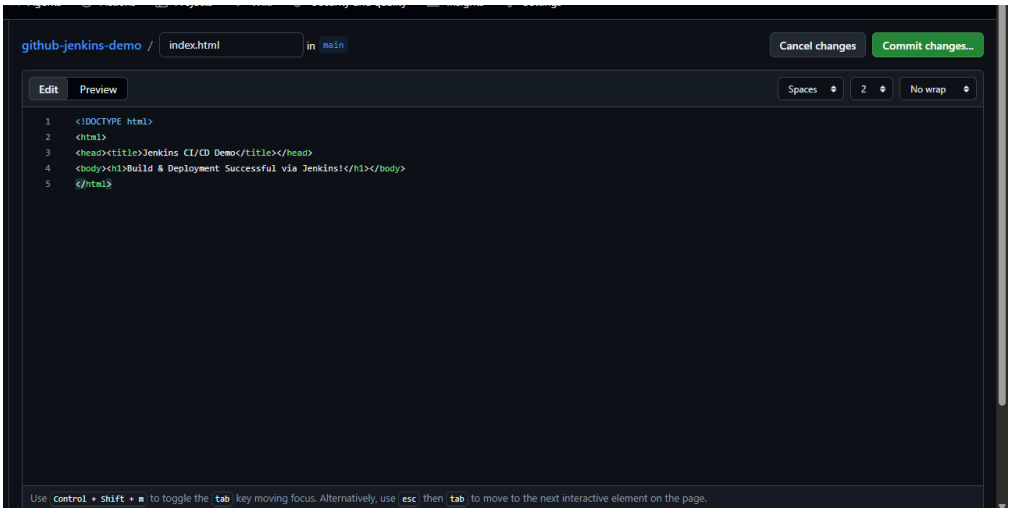

localhost:8081/job/GitHub-Jenkins-Demo1/configure
Verify it's you
All Bookmarks
Jenkins / GitHub-Jenkins-Demo1 / Configure
Configure
General
Triggers
Pipeline
Advanced
Pipeline
Define your Pipeline using Groovy directly or pull it from source control.
Definition
Pipeline script
Script ?
try sample Pipeline...
pipeline {
agent any
stages {
stage('Build') {
steps {
echo 'Building the application...'
}
}
stage('Test') {
steps {
echo 'Testing the application...'
}
}
stage('Deploy') {
steps {
Use Groovy Sandbox ?
Save
Apply

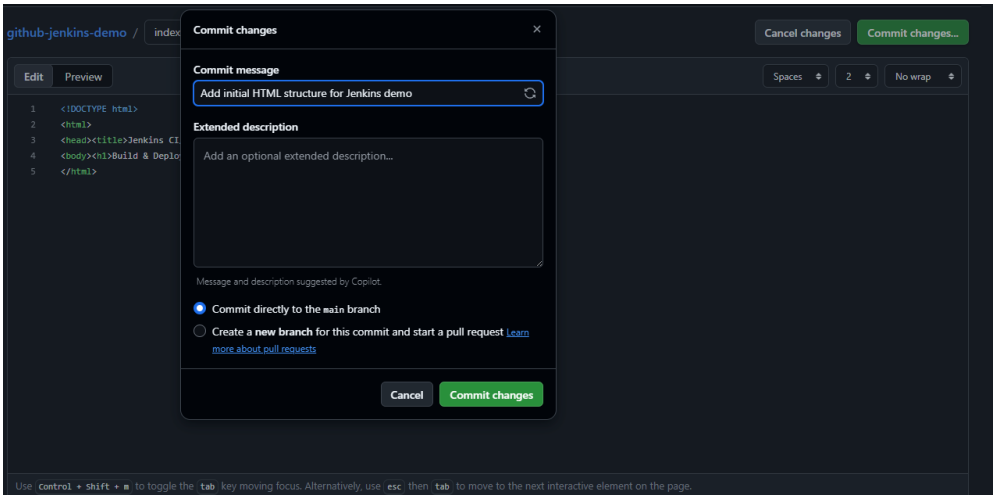

localhost:8081/job/GitHub-Jenkins-Demo1/
Verify it's you
All Bookmarks
Jenkins / GitHub-Jenkins-Demo1
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
Builds
No builds
GitHub-Jenkins-Demo1
Add description
Permalinks
REST API

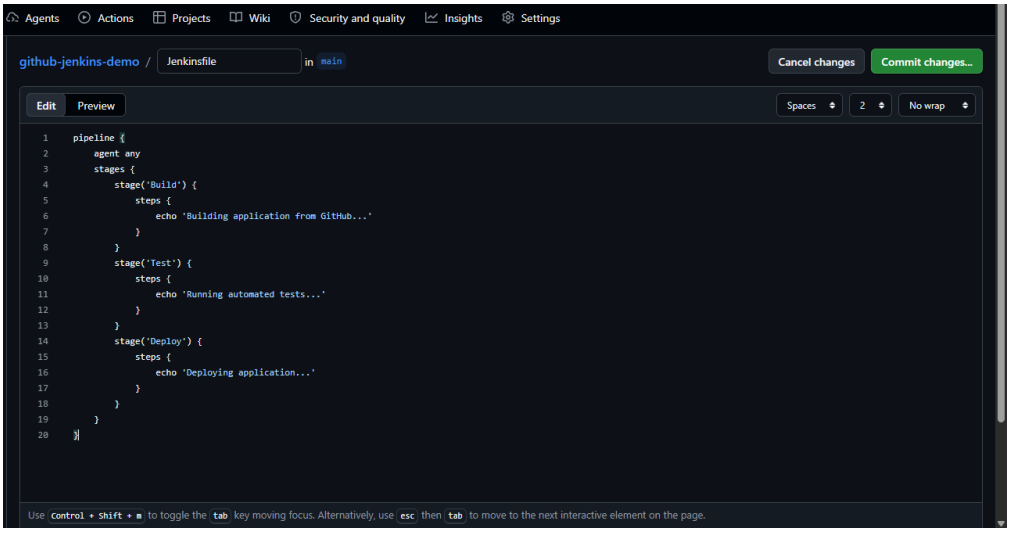

localhost:8081/job/GitHub-Jenkins-Demo1/
Verify it's you
All Bookmarks
Jenkins
GitHub-Jenkins-Demo1
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
GitHub-Jenkins-Demo1
Add description
Permalinks
Builds
No builds
Build scheduled
REST API
Jenkins 2.568.2

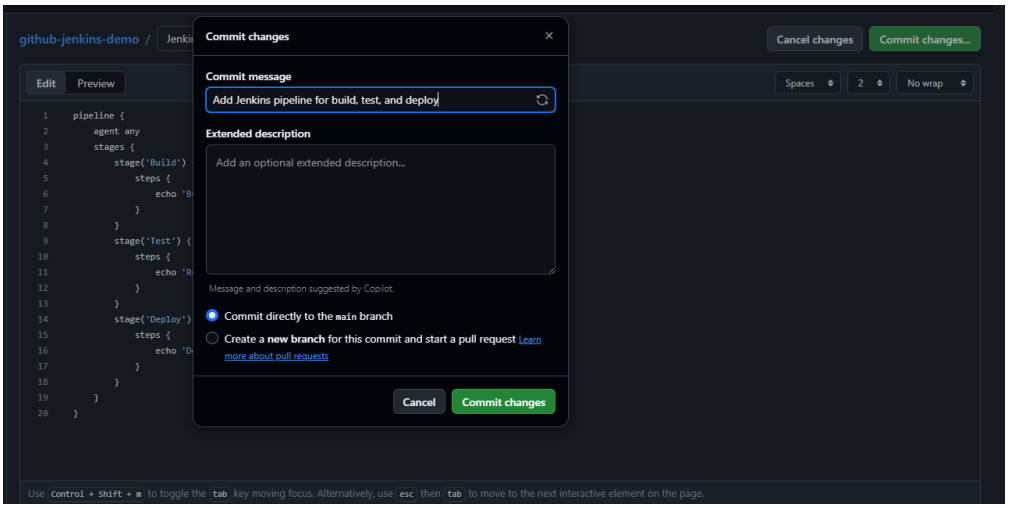

localhost:8081/job/GitHub-Jenkins-Demo1/
Verify it's you
All Bookmarks
Jenkins
GitHub-Jenkins-Demo1
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
GitHub-Jenkins-Demo1
Add description
Permalinks
Changes
Console Output
Delete build '#1'
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Builds
Today
#1 3:07 PM
REST API
Jenkins 2.568.2


localhost:8081/job/GitHub-Jenkins-Demo1/1/console
Jenkins
GitHub-Jenkins-Demo1
#1
Console Output
Delete build '#1'
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Started by user Soham Ajwani
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\GitHub-Jenkins-Demo1
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Building the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Testing the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS

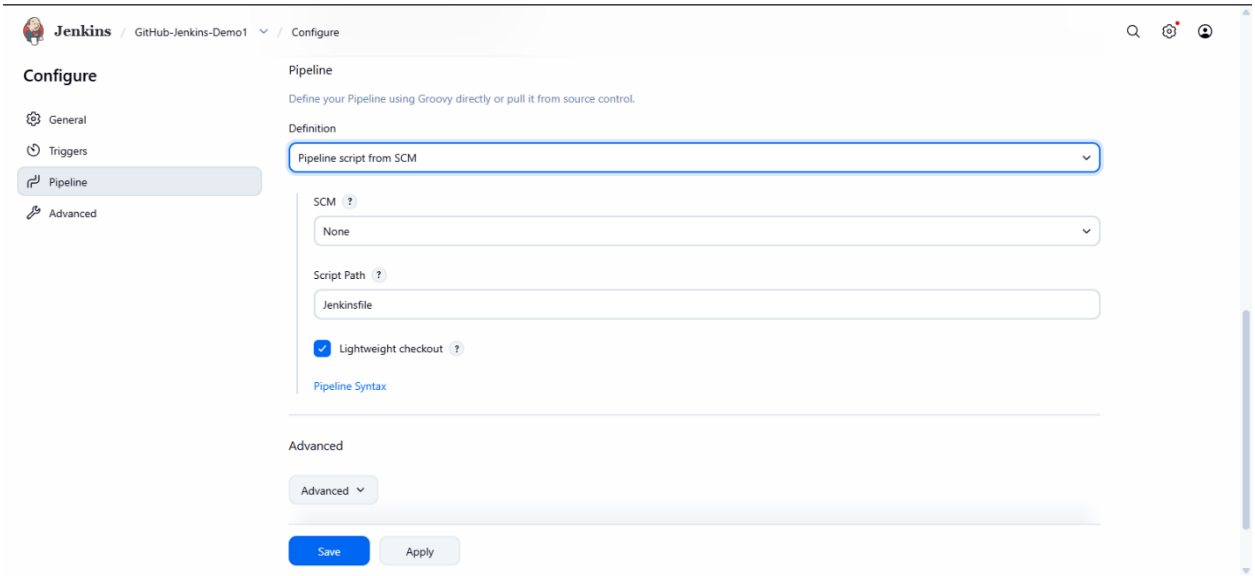

github-jenkins-demo / index.html in main
Cancel changes
Commit changes...
Edit
Preview
Spaces
2
No wrap
<!DOCTYPE html>
<html>
<head><title>Jenkins CI/CD Demo</title></head>
<body><h1>Build & Deployment Successful via Jenkins!</h1></body>
</html>
Use Control + Shift + m to toggle the tab key moving focus. Alternatively, use esc then tab to move to the next interactive element on the page.

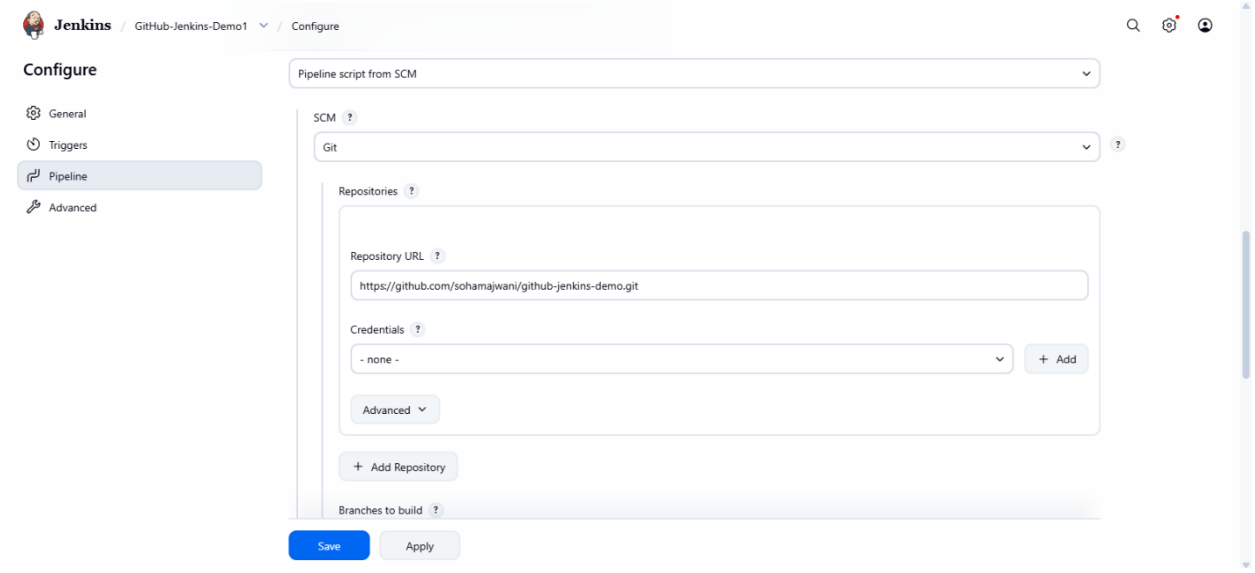

github-jenkins-demo /
Cancel changes
Commit changes...
Edit
Preview
Commit changes
Commit message
Add initial HTML structure for Jenkins demo
Extended description
Add an optional extended description...
Message and description suggested by Copilot.
Commit directly to the main branch
Create a new branch for this commit and start a pull request Learn more about pull requests
Cancel
Commit changes

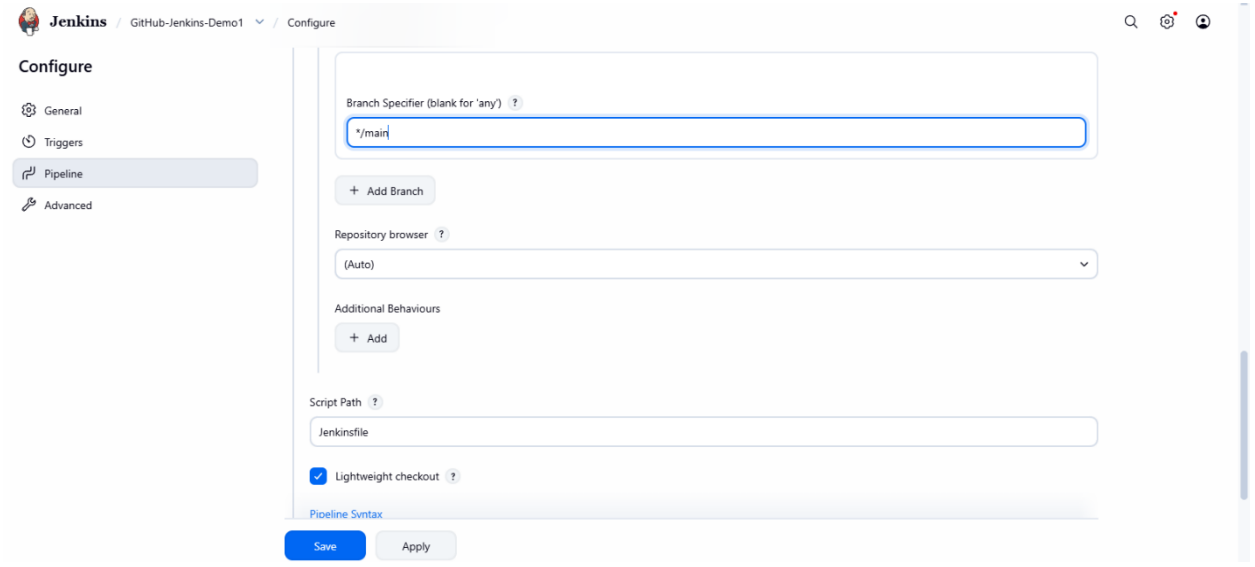

Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
github-jenkins-demo / Jenkinsfile in main
Cancel changes
Commit changes...
Edit
Preview
Spaces
2
No wrap
pipeline {
agent any
stages {
stage('Build') {
steps {
echo 'Building application from GitHub...'
}
}
stage('Test') {
steps {
echo 'Running automated tests...'
}
}
stage('Deploy') {
steps {
echo 'Deploying application...'
}
}
}
}
Use Control + Shift + m to toggle the tab key moving focus. Alternatively, use esc then tab to move to the next interactive element on the page.

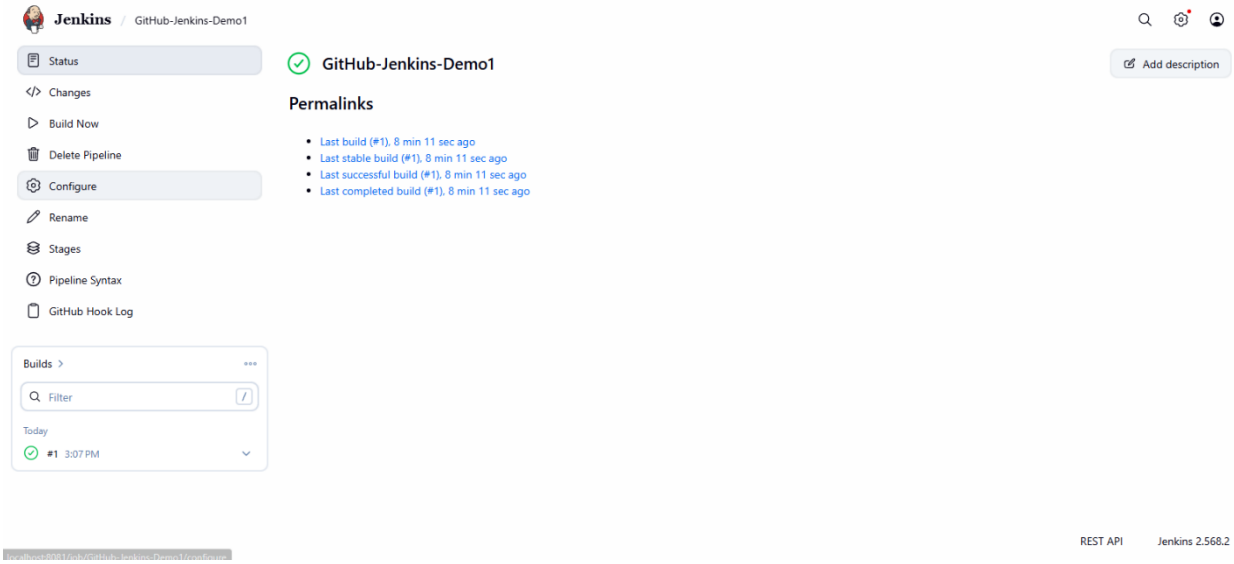

github-jenkins-demo /
Commit changes
Commit message
Add Jenkins pipeline for build, test, and deploy
Extended description
Add an optional extended description...
Message and description suggested by Copilot.
Commit directly to the main branch
Create a new branch for this commit and start a pull request Learn more about pull requests
Cancel
Commit changes
Cancel changes
Commit changes...
Edit
Preview
Spaces
2
No wrap
Use Control + Shift + m to toggle the tab key moving focus. Alternatively, use esc then tab to move to the next interactive element on the page.

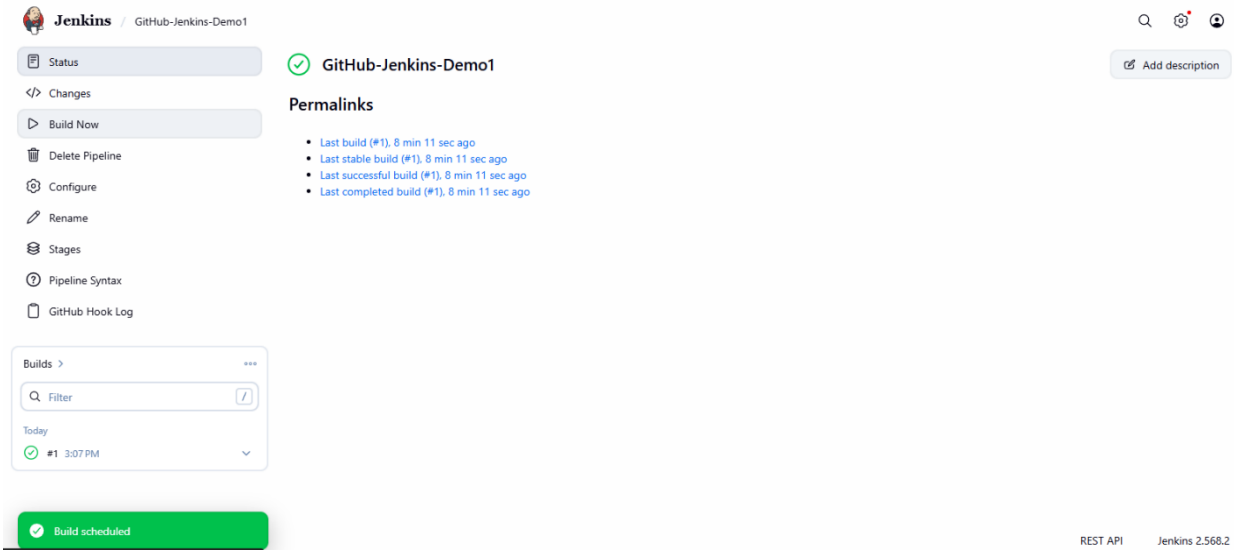

Jenkins
GitHub-Jenkins-Demo1
Configure
Configure
General
Triggers
Pipeline
Advanced
Triggers
Set up automated actions that start your build based on specific events, like code changes or scheduled times.
Build after other projects are built
Build periodically
GitHub hook trigger for GITScm polling
Poll SCM
Trigger builds remotely (e.g., from scripts)
Pipeline
Define your Pipeline using Groovy directly or pull it from source control.
Definition
Pipeline script
Script
pipeline {
agent any
stages {
stage('Build') {
Save
Apply

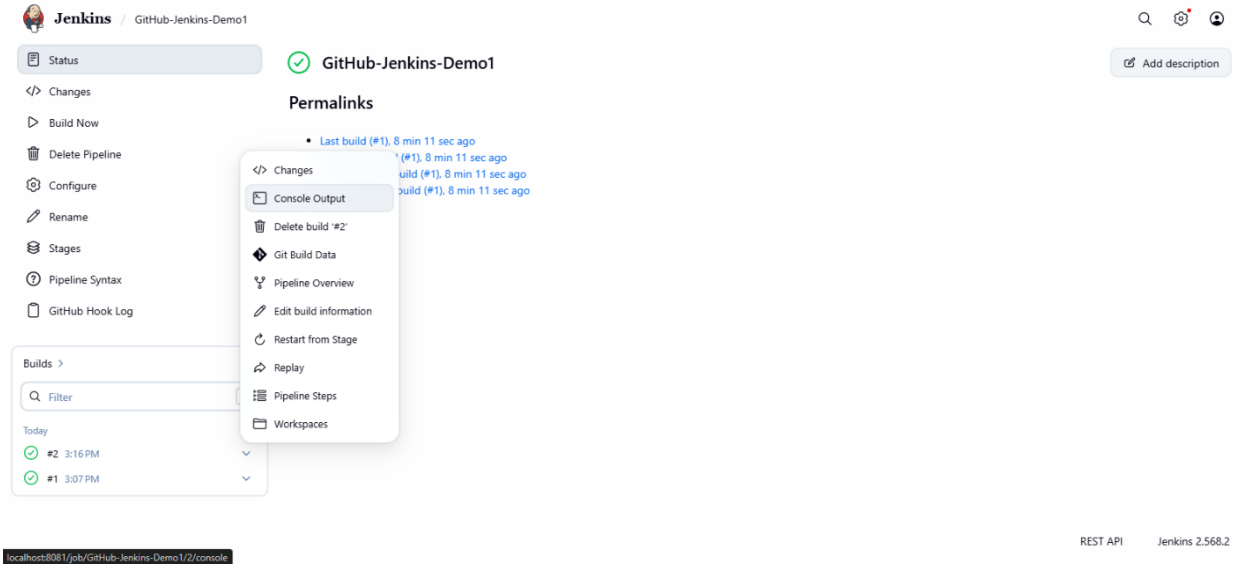

Jenkins / GitHub-Jenkins-Demo1 / Configure
Configure
General
Triggers
Pipeline
Advanced
Pipeline
Define your Pipeline using Groovy directly or pull it from source control.
Definition
Pipeline script from SCM
SCM
None
Script Path
Jenkinsfile
Lightweight checkout
Pipeline Syntax
Advanced
Advanced
Save
Apply

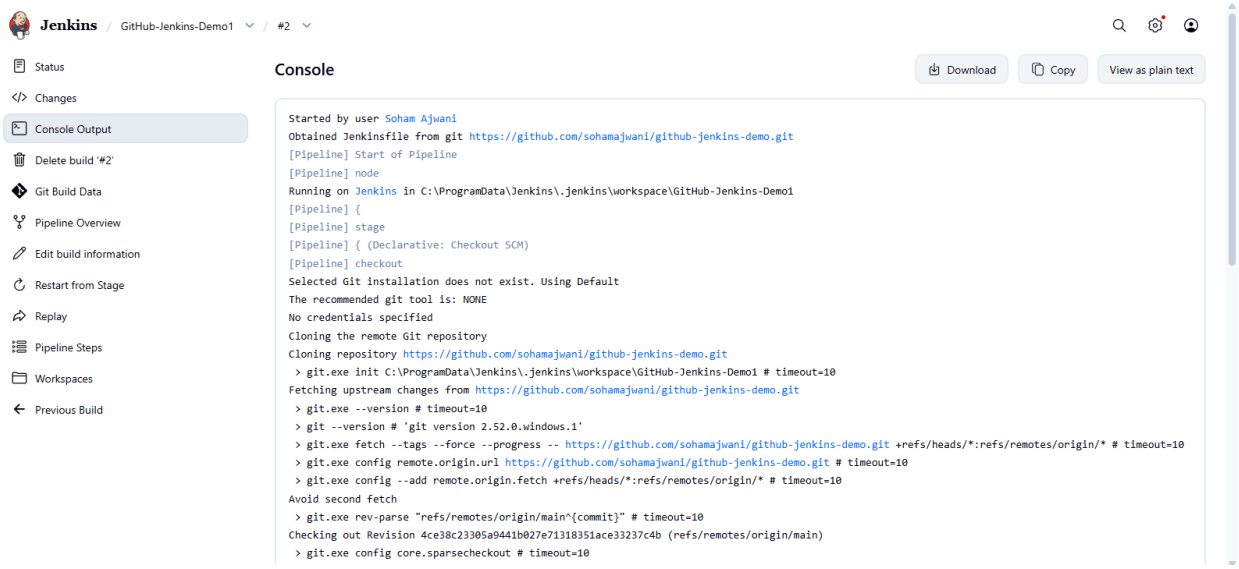

Jenkins / GitHub-Jenkins-Demo1 / Configure
Configure
General
Triggers
Pipeline
Advanced
Pipeline script from SCM
SCM
Git
Repositories
Repository URL
https://github.com/sohamajwani/github-jenkins-demo.git
Credentials
- none -
+ Add
Advanced
+ Add Repository
Branches to build
Save
Apply

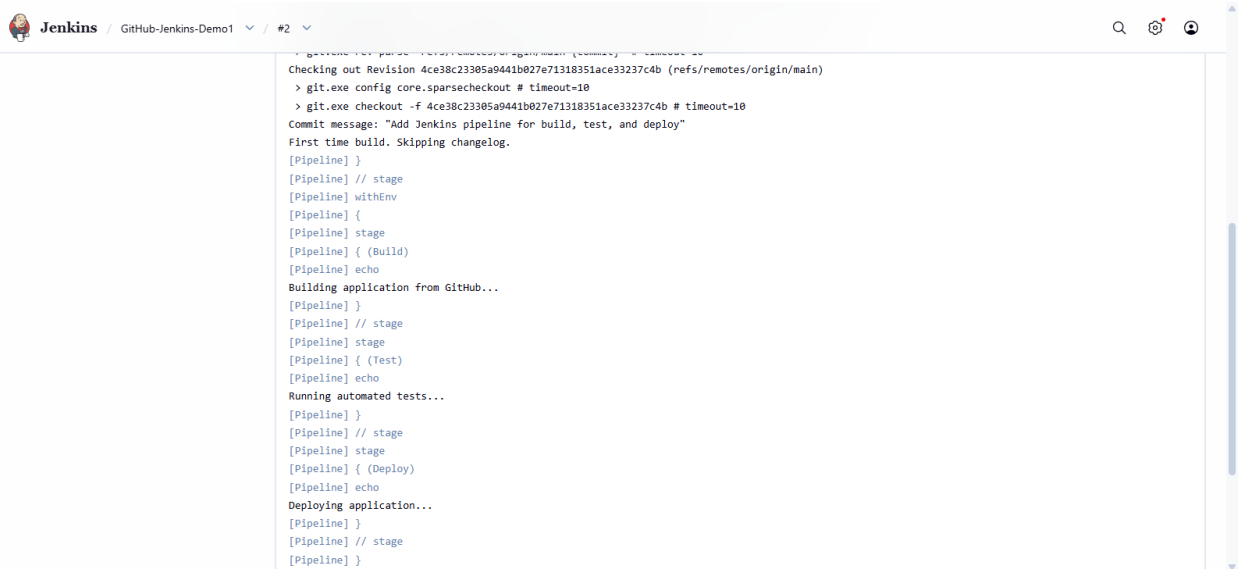

Jenkins / GitHub-Jenkins-Demo1 / Configure
Configure
General
Triggers
Pipeline
Advanced
Branch Specifier (blank for 'any')
*/main
+ Add Branch
Repository browser
(Auto)
Additional Behaviours
+ Add
Script Path
Jenkinsfile
Lightweight checkout
Pipeline Syntax
Save
Apply


Jenkins / GitHub-Jenkins-Demo1
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
GitHub Hook Log
Builds
Filter
Today
#1 3:07 PM
GitHub-Jenkins-Demo1
Add description
Permalinks
Last build (#1), 8 min 11 sec ago
Last stable build (#1), 8 min 11 sec ago
Last successful build (#1), 8 min 11 sec ago
Last completed build (#1), 8 min 11 sec ago
REST API
Jenkins 2.568.2


Jenkins / GitHub-Jenkins-Demo1
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
GitHub Hook Log
Builds
Filter
Today
#1 3:07 PM
Build scheduled
GitHub-Jenkins-Demo1
Add description
Permalinks
Last build (#1), 8 min 11 sec ago
Last stable build (#1), 8 min 11 sec ago
Last successful build (#1), 8 min 11 sec ago
Last completed build (#1), 8 min 11 sec ago
REST API
Jenkins 2.568.2

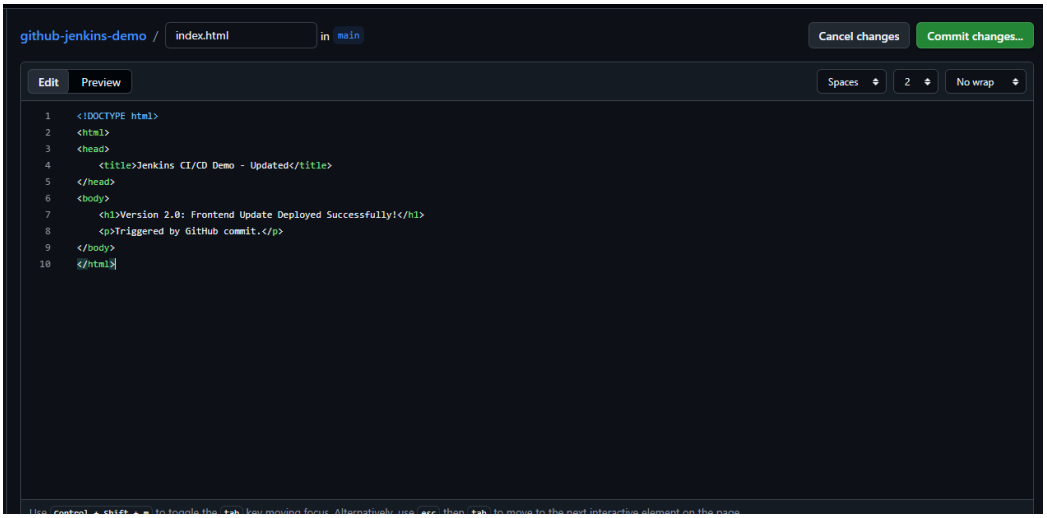

Jenkins / GitHub-Jenkins-Demo1
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
GitHub Hook Log
Builds
Filter
Today
#2 3:16 PM
#1 3:07 PM
Changes
Console Output
Delete build '#2'
Git Build Data
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
GitHub-Jenkins-Demo1
Add description
Permalinks
Last build (#1), 8 min 11 sec ago
REST API
Jenkins 2.568.2
localhost:8081/job/GitHub-Jenkins-Demo1/2/console

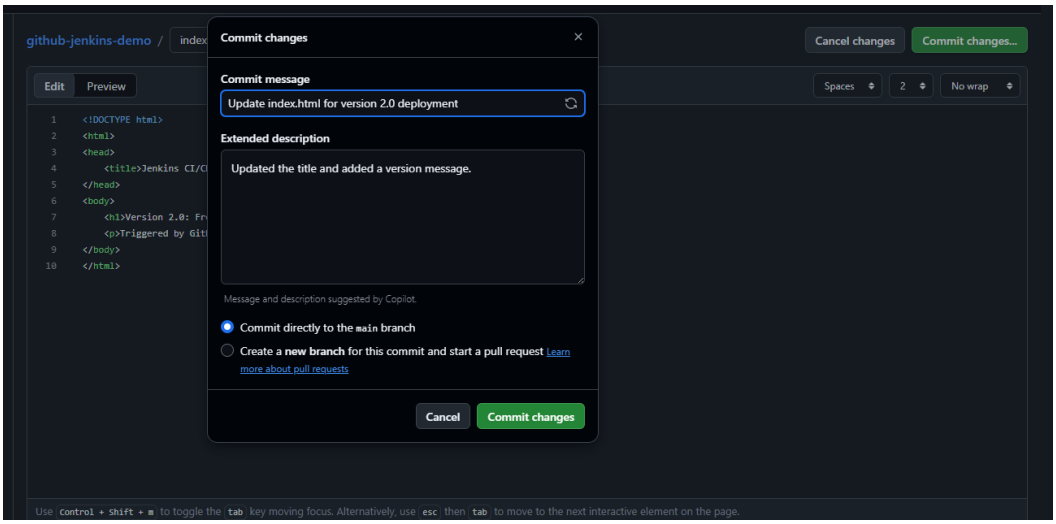


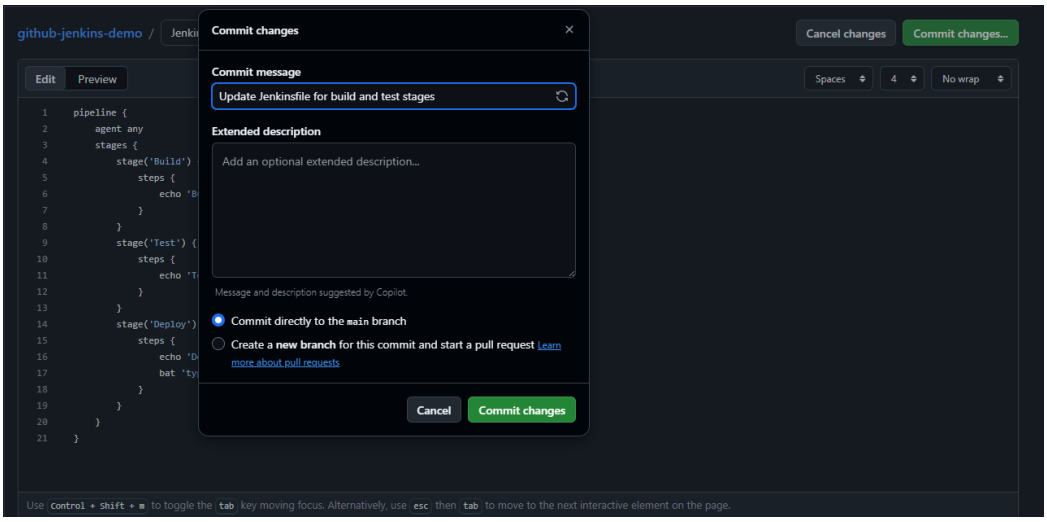


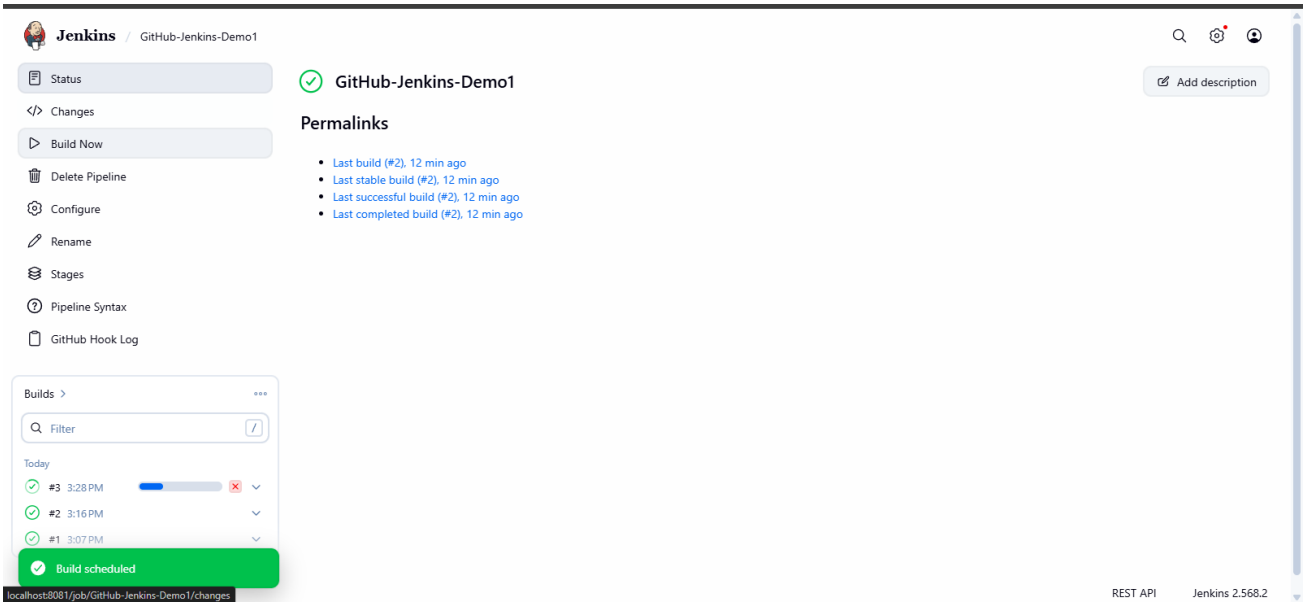


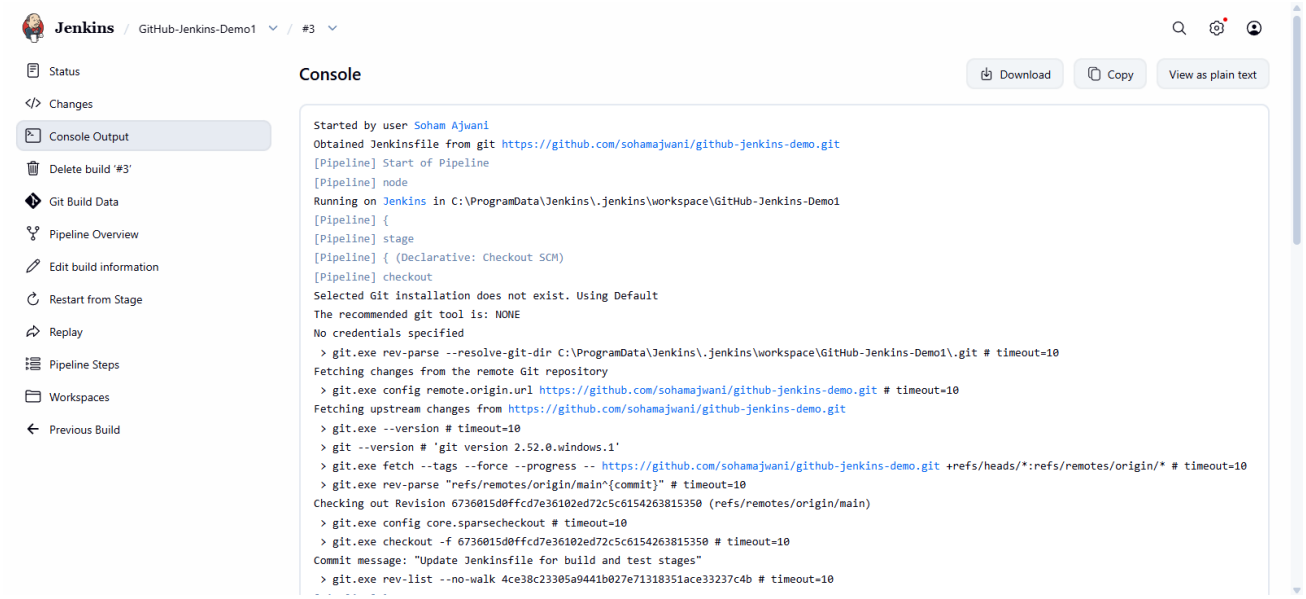
Jenkins 2.568.2


github-jenkins-demo / index.html in main
Cancel changes
Commit changes...
Edit
Preview
Spaces
2
No wrap
<!DOCTYPE html>
<html>
<head>
<title>Jenkins CI/CD Demo - Updated</title>
</head>
<body>
<h1>Version 2.0: Frontend Update Deployed Successfully!</h1>
Triggered by GitHub commit.
</body>
</html>


github-jenkins-demo /
Commit changes
Commit message
Update index.html for version 2.0 deployment
Extended description
Updated the title and added a version message.
Message and description suggested by Copilot.
Commit directly to the main branch
Create a new branch for this commit and start a pull request Learn more about pull requests
Cancel
Commit changes
Cancel changes
Commit changes...
Edit
Preview
Spaces
2
No wrap
Use Control + Shift + m to toggle the tab key moving focus. Alternatively, use esc then tab to move to the next interactive element on the page.


github-jenkins-demo /
Commit changes
Commit message
Update Jenkinsfile for build and test stages
Extended description
Add an optional extended description...
Message and description suggested by Copilot.
Commit directly to the main branch
Create a new branch for this commit and start a pull request Learn more about pull requests
Cancel
Commit changes
Cancel changes
Commit changes...
Edit
Preview
Spaces
4
No wrap
pipeline {
agent any
stages {
stage('Build')
steps {
}
}
stage('Test') {
steps {
}
}
stage('Deploy')
steps {
}
}
}
}
Use Control + Shift + m to toggle the tab key moving focus. Alternatively, use esc then tab to move to the next interactive element on the page.


Jenkins / GitHub-Jenkins-Demo1
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
GitHub Hook Log
Builds
Filter
Today
#3 3:28 PM
#2 3:16 PM
#1 3:07 PM
Build scheduled
localhost:8081/job/GitHub-Jenkins-Demo1/changes
GitHub-Jenkins-Demo1
Add description
Permalinks
Last build (#2), 12 min ago
Last stable build (#2), 12 min ago
Last successful build (#2), 12 min ago
Last completed build (#2), 12 min ago
REST API
Jenkins 2.568.2


Jenkins / GitHub-Jenkins-Demo1 / #3
Status
Changes
Console Output
Delete build '#3'
Git Build Data
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
Console
Download
Copy
View as plain text
Started by user Soham Ajwani
Obtained Jenkinsfile from git https://github.com/sohamajwani/github-jenkins-demo.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\GitHub-Jenkins-Demo1
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
> git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\.jenkins\workspace\GitHub-Jenkins-Demo1\.git # timeout=10
Fetching changes from the remote Git repository
> git.exe config remote.origin.url https://github.com/sohamajwani/github-jenkins-demo.git # timeout=10
Fetching upstream changes from https://github.com/sohamajwani/github-jenkins-demo.git
> git.exe --version # timeout=10
> git --version # 'git version 2.52.0.windows.1'
> git.exe fetch --tags --force --progress -- https://github.com/sohamajwani/github-jenkins-demo.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 6736015d0ffcd7e36102ed72c5c6154263815350 (refs/remotes/origin/main)
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f 6736015d0ffcd7e36102ed72c5c6154263815350 # timeout=10
Commit message: "Update Jenkinsfile for build and test stages"
> git.exe rev-list --no-walk 4ce38c23305a9441b027e71318351ace33237c4b # timeout=10


Jenkins / GitHub-Jenkins-Demo1 / #3
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying application...
[Pipeline] bat
C:\ProgramData\Jenkins\.jenkins\workspace\GitHub-Jenkins-Demo1>type index.html
<!DOCTYPE html>
<html>
<head>
<title>Jenkins CI/CD Demo - Updated</title>
</head>
<body>
<h1>Version 2.0: Frontend Update Deployed Successfully!</h1>
Triggered by GitHub commit.
</body>
</html>
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS



Version 2.0: Frontend Update Deployed Successfully!

Triggered by GitHub commit.